

Problem 9.1 A bar is subjected to a uniform tensile stress of 200 N/mm². Find the intensity of shear stress, normal stress and resultant stress on a plane the normal to which is inclined 30° to the axis of the bar.

Solution: $\sigma_n = \sigma \cos^2 \theta = 200 \cos^2 30^\circ = 150 \text{ N/mm}^2$

$$\sigma_t = \frac{\sigma}{2} \sin 2\theta = 100 \sin 60^\circ = 86.6 \text{ N/mm}^2$$

$$\begin{aligned}\sigma_r &= \sqrt{\sigma_n^2 + \sigma_t^2} = \sqrt{150^2 + 86.6^2} \\ &= 173.2 \text{ N/mm}^2\end{aligned}$$

Problem 9.2 A piece of material is subjected to tensile stress of 140 N/mm² and 100 N/mm² at right angles to each other. Find fully the stresses on a plane the normal of which makes an angle of 40° with the 140 N/mm² stress.

Solution:

From equation 9.7 we have

$$\begin{aligned}\sigma_n &= \sigma_1 \cos^2 \theta + \sigma_2 \sin^2 \theta \\ &= 140 \cos^2 40^\circ + 100 \sin^2 40^\circ \\ &= 126.84 \text{ N/mm}^2\end{aligned}$$

Similarly, from equation 9.9, we have

$$\begin{aligned}\sigma_t &= \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta \\ &= \frac{140 - 100}{2} \sin 80^\circ = \\ &= 18.8 \text{ N/mm}^2\end{aligned}$$

The resultant stress $\sigma_r = \sqrt{(126.84)^2 + (18.8)^2}$
 $= 128.22 \text{ N/mm}^2$

Problem 9.3 A piece of material is subjected to tensile stress of 140 N/mm² in one direction and a compressive stress of 100 N/mm² in a direction at right angles to the previous one. Find fully the stresses on a plane the normal of which makes an angle of 40° with the 140 N/mm² stress.

Solution:

In this case, σ_1 is tensile and σ_2 is compressive.

Hence, From equation 9.7 we have

$$\begin{aligned}\sigma_n &= \sigma_1 \cos^2 \theta + \sigma_2 \sin^2 \theta \\ &= 140 \cos^2 40^\circ - 100 \sin^2 40^\circ \\ &= 40.84 \text{ N/mm}^2.\end{aligned}$$

$$\begin{aligned}\text{and } \sigma_t &= \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta \\ &= \frac{140 + 100}{2} \sin 80^\circ = 118.18 \text{ N/mm}^2 \\ \sigma_r &= \sqrt{\sigma_n^2 + \sigma_t^2} = \sqrt{(40.84)^2 + (118.18)^2} \\ &= 125.04 \text{ N/mm}^2\end{aligned}$$

Problem 9.4 The principal tensile stresses at a point across two perpendicular planes are 160 N/mm^2 and 80 N/mm^2 . Find the normal and tangential stress and the resultant stress and its obliquity on a plane at 20° with the major principal plane.

Solution: On a plane at 20° with the major principal plane, from equation (9.8)

$$\begin{aligned}\sigma_n &= \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta \\ &= \frac{160 + 80}{2} + \frac{160 - 80}{2} \cos 40^\circ \\ &= 120 + 40 \cos 40^\circ = 150.64 \text{ N/mm}^2 \text{ (tensile)}\end{aligned}$$

$$\begin{aligned}\text{from equation (9.9) } \sigma_t &= \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta \\ \sigma_t &= \frac{160 - 80}{2} \sin 40^\circ \\ &= 40 \sin 40^\circ \\ &= 25.72 \text{ N/mm}^2\end{aligned}$$

$$\therefore \text{ Resultant stress} = \sqrt{150.64^2 + 25.72^2} = 152.82 \text{ N/mm}^2$$

$$\text{Obliquity } \phi = \tan^{-1} \frac{\sigma_t}{\sigma_n} = \tan^{-1} \frac{25.72}{150.64}$$

$$\therefore \phi = 9^\circ 41'$$

Problem 9.5 The principal stresses at a point in a bar are 400 N/mm^2 (tensile) and 200 N/mm^2 (compressive). Determine the resultant stress in magnitude and direction on a

plane inclined at 60° to the axis of the major principal stress. Also determine the maximum intensity of shear stress in the material at the point.

Solution:

$$\sigma_1 = 400 \text{ N/mm}^2 \text{ (tensile)}$$

$$\sigma_2 = -200 \text{ N/mm}^2 \text{ (compressive)}$$

$$\theta = 90^\circ - 60^\circ = 30^\circ$$

$$\begin{aligned} \text{Normal stress on the plane } \sigma_n &= \frac{\sigma_1 + \sigma_2}{2} + \frac{\sigma_1 - \sigma_2}{2} \cos 2\theta \\ &= 100 + 300 \times \frac{1}{2} = 250 \text{ N/mm}^2 \end{aligned}$$

$$\sigma_t = \frac{\sigma_1 - \sigma_2}{2} \sin 2\theta = \frac{400 + 200}{2} \sin 60^\circ = 259.8 \text{ N/mm}^2$$

$$\text{Resultant stress } \sigma = \sqrt{\sigma_n^2 + \sigma_t^2} = \sqrt{250^2 + 259.8^2} = 360.54 \text{ N/mm}^2$$

$$\tan \phi = \frac{\sigma_t}{\sigma_n} = \frac{259.8}{250} \quad \therefore \theta = 46^\circ.1$$

$$\text{Maximum shear stress } \tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{400 + 200}{2} = 300 \text{ N/mm}^2$$

Problem 9.6 At a certain point in a strained material the principal stresses are 200 N/mm^2 and 80 N/mm^2 both tensile. Find the normal, tangential and resultant stresses across a plane through the point at 48° to the major principal plane, using Mohr's circle of stress.

Solution: Draw OA and OB to represent the two principal stresses.

$$\therefore \quad \text{OA} = \sigma_1 = 200 \text{ N/mm}^2$$

$$\text{OB} = \sigma_2 = 80 \text{ N/mm}^2$$

On AB as diameter describe a circle draw BD so that

$$\text{DBA} = \theta = 48^\circ$$

Draw DE perpendicular to OA.

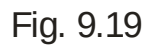
Now OE represents normal stress on the given plane. By measurement, $\text{OE} = 133.8 \text{ N/mm}^2$. QP represents the tangential stress on the given plane. By measurement, $\text{ED} = 59.6 \text{ N/mm}^2$.

$$\sigma_n = \text{OE} = 133.8 \text{ N/mm}^2$$

$$\sigma_t = \text{ED} = 59.6 \text{ N/mm}^2$$

$$\phi = \text{DOB} = 24^\circ$$

$$\alpha = \text{ODB} = 24^\circ$$



Resultant stress is given by OD.

Inclination of the resultant stress with the direction of the major principal stress

$$\alpha = \overset{\wedge}{O D B} = \theta - \phi = 48^0 - 24^0 = 24^0.$$

Fig. 9.19

Solution: Fig. 9.19 shows the Mohr's circle corresponding to the given principal stresses.

$$\begin{aligned} OA &= \sigma_1 = 160 \text{ N/mm}^2 \text{ (tensile)} \\ OB &= \sigma_2 = 100 \text{ N/mm}^2 \text{ (compressive)} \\ OE &= \sigma_{n1} = 123.6 \text{ N/mm}^2 \text{ (tensile)} \\ DE &= \sigma_{t1} = 90.4 \text{ N/mm}^2 \\ OG &= \sigma_{n2} = 50 \text{ N/mm}^2 \text{ (compressive)} \\ FG &= \sigma_{t2} = 102.4 \text{ N/mm}^2 \\ BFO &= \alpha_1 = 14^\circ 11' \\ BFO &= \alpha_2 = 52^\circ 2' \end{aligned}$$

Since σ_1 and σ_2 are unlike OA and OB have been drawn on opposite sides of O, D and F correspond to the 22° and 64° planes. The results are given in the figure.

Problem 9.8 A rectangular block of material is subjected to a tensile stress of 220 N/mm^2 on one plane and a tensile stress of 94 N/mm^2 on a plane at right angle, together with shear stresses of 126 N/mm^2 on the same planes. Find:

- (i) The direction of the principal planes.
- (ii) The magnitude of the principal stress.
- (iii) The magnitude of the greatest shear stress.

Solution: Let σ_1 and σ_2 be the principle stresses.

The inclination of the principal planes with the plane AB carrying the tensile stress of $\sigma = 220 \text{ N/mm}^2$ is given by

$$\tan 2\theta = \frac{2\tau}{\sigma_1 - \sigma_2} = \frac{2 \times 126}{220 - 94}$$

$$\begin{aligned} \therefore 2\theta &= 63^\circ 26' \text{ or } 243^\circ 26' \\ \theta &= 31^\circ 43' \text{ or } 121^\circ 43' \end{aligned}$$

$$\begin{aligned} \text{Major principal stress} \quad &= \sigma_{np1} = \frac{\sigma_1 + \sigma_2}{2} + \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2} \\ &= \frac{220 + 94}{2} + \sqrt{\left(\frac{220 - 94}{2}\right)^2 + 126^2} \text{ N/mm}^2 \\ &= 157 + 140.8 = 297.8 \text{ N/mm}^2 \text{ (tensile)} \end{aligned}$$

$$\begin{aligned} \text{Major principal stress} \quad &= \sigma_{np2} = \frac{\sigma_1 + \sigma_2}{2} - \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2} \\ &= 157 - 140.8 = 16.2 \text{ N/mm}^2 \text{ (tensile)} \end{aligned}$$

Maximum shear stress $\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{297.8 - 16.2}{2} = 140.8 \text{ N/mm}^2$

This will occur on planes at $31^\circ 43' + 45^\circ = 76^\circ 43'$

and $76^\circ 43' + 90^\circ = 166^\circ 43'$ with the plane AB carrying the normal stress of 220 N/mm^2 .

Problem 9.9 At a certain point in a strained material the intensities of normal stresses on two planes at right angles to each other are 40 N/mm^2 and 20 N/mm^2 both tensile. They are accompanied by shear stress of 20 N/mm^2 . find the principal planes and the principal stresses. Find also the maximum shear stress.

Solution: $\sigma_1 = 40 \text{ N/mm}^2$; $\sigma_2 = 20 \text{ N/mm}^2$; $\tau = 20 \text{ N/mm}^2$

For the position of the principal planes,

$$\tan 2\theta = \frac{2\tau}{\sigma_1 - \sigma_2} = \frac{2 \times 20}{40 - 20} = 2 \therefore 2\theta = 63^\circ 26' ; 243^\circ 26'$$

$$\theta = 31^\circ 43' ; 121^\circ 43'$$

$$\frac{\sigma_1 - \sigma_2}{2} = \frac{40 - 20}{2} = 10 \text{ N/mm}^2 \quad \frac{P - P}{2} = \frac{40 - 20}{2} = 10 \text{ N/mm}^2 ;$$

The principal stresses are given by

$$\begin{aligned} \sigma_{np1} &= \frac{\sigma_1 + \sigma_2}{2} \pm \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2} \\ &= 30 \pm \sqrt{10^2 + 20^2} = 30 \pm 22.36 \text{ N/mm}^2 \end{aligned}$$

$$\therefore P_1 = 30 + 22.36 = 52.36 \text{ N/mm}^2$$

$$P_2 = 30 - 22.36 = 7.64 \text{ N/mm}^2$$

$$\begin{aligned} \text{Maximum shear stress } \tau_{\max} &= \frac{\sigma_1 - \sigma_2}{2} = \frac{52.36 - 7.64}{2} = 22.36 \text{ N/mm}^2 \\ &= 22.36 \text{ N/mm}^2 \end{aligned}$$

Problem 9.10 A rectangular block of material is subjected to a tensile stress of 200 N/mm^2 on one plane and a tensile stress of 100 N/mm^2 on a plane at right angles, together with shear stresses of 120 N/mm^2 on the same planes. Find :

- (i) The direction of the principal planes.
- (ii) The magnitude of the principal stresses.
- (iii) The magnitude of the greatest shear stress.

Solution: Let σ_1 and σ_2 be the principal stresses.

Let θ be the inclination of a principal plane with the plane carrying the normal tensile stress of 100 N/mm^2 .

$$\tan 2\theta = \frac{2\tau}{\sigma_1 - \sigma_2} = \frac{2 \times 120}{200 - 100} = 2.40$$

$$\therefore \quad \begin{aligned} 2\theta &= 67^\circ 22' \quad \text{or} \quad 247^\circ 22' \\ \theta &= 33^\circ 41' \quad \text{or} \quad 123^\circ 41' \end{aligned}$$

$$\begin{aligned} \text{Major principal stress} \quad &= \sigma_{np1} = \frac{\sigma_1 + \sigma_2}{2} + \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2} \\ &= \frac{200 + 100}{2} + \sqrt{\left(\frac{200 - 100}{2}\right)^2 + 120^2} \\ &= 150 + 130 = 280 \text{ N/mm}^2 \text{ (tensile)} \\ &= \sigma_{np1} = \frac{\sigma_1 + \sigma_2}{2} - \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2} \\ &= 150 - 130 = 20 \text{ N/mm}^2 \text{ (tensile)} \end{aligned}$$

$$\text{Maximum shear stress} = \frac{\sigma_1 - \sigma_2}{2} = \frac{280 - 20}{2} = 130 \text{ N/mm}^2$$

The maximum shear stress will occur on planes at $33^\circ 41' + 45^\circ = 78^\circ 41'$, and $78^\circ 41' + 90^\circ = 168^\circ 41'$ with the plane AB carrying the normal stress of 100 N/mm^2 .

Problem 9.11 Direct stresses of 240 N/mm^2 (tensile) and 180 N/mm^2 (compressive) exist on two perpendicular planes at a certain point in a body. They are also accompanied by shear stresses on the planes. The greater principal stress at the point due to these is 300 N/mm^2 .

(a) Find the shear stresses on these planes

(b) Find also the maximum shear stress at the point

Solution: $\sigma_1 = 240 \text{ N/mm}^2 \quad \sigma_2 = -180 \text{ N/mm}^2$
 $\sigma_{np1} = 300 \text{ N/mm}^2$.

The sum of the normal stresses on two mutually perpendicular planes is constant

$$\begin{aligned} \therefore \quad &\sigma_{np1} + \sigma_{np2} = \sigma_1 + \sigma_2 \\ &300 + \sigma_{np2} = 240 - 180 \\ \therefore \quad &\sigma_{np2} = -240 \text{ N/mm}^2 \text{ (compressive)} \end{aligned}$$

Also
$$\sigma_{np1} = \frac{\sigma_1 + \sigma_2}{2} + \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2} = 300 \text{ N/mm}^2$$

$$\therefore \quad \frac{240 - 180}{2} + \sqrt{\left(\frac{240 + 180}{2}\right)^2 + \tau^2} = 300$$

$$15 + \sqrt{44100 + \tau^2} = 300$$

$$44100 + \tau^2 = (300 - 15)^2 = 72900$$

$$\tau^2 = 28800$$

$$\therefore \tau = 169.70 \text{ N/mm}^2.$$

$$\text{Maximum shear stress } \tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{300 - 240}{2} \text{ N/mm}^2 = 270 \text{ N/mm}^2.$$

Problem 9.12 Two planes AB and BC which are at right angles carry shear stresses of intensity 35 N/mm^2 while these planes also carry a tensile stress of 140 N/mm^2 and a compressive stress of 70 N/mm^2 respectively.

Determine the principal planes and the principal stresses. Also determine the maximum shear stress and the planes on which it acts.

Solution:

$$\sigma_1 = 140 \text{ N/mm}^2$$

$$\sigma_2 = -70 \text{ N/mm}^2$$

$$\tau = 35 \text{ N/mm}^2$$

Principal stresses are given by

$$\frac{\sigma_1 + \sigma_2}{2} \pm \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2}$$

$$= \frac{140 + 70}{2} \pm \sqrt{\left(\frac{140 - 70}{2}\right)^2 + 35^2}$$

$$= 35 \pm 110.68 \text{ N/mm}^2.$$

$$\sigma_{np1} = 17.5 + 55.34 = 145.68 \text{ N/mm}^2$$

$$\sigma_{np2} = 17.5 - 55.34 = -75.68 \text{ N/mm}^2$$

Positions of the principal planes

$$\tan 2\theta = \frac{2\tau}{\sigma_1 - \sigma_2} = \frac{2 \times 35}{140 - 61} = 0.3333$$

$$2\theta = 18^\circ.43 \text{ and } 198.43^\circ$$

$$\therefore \theta = 9^\circ.215 \text{ and } 99^\circ.215$$

$$\text{i.e., } \theta_1 = 9^\circ.215 \text{ and } \theta_2 = 99^\circ.215$$

Maximum shear stress

$$\tau_{\max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{145.68 - 75.68}{2} = 110.68 \text{ N/mm}^2$$

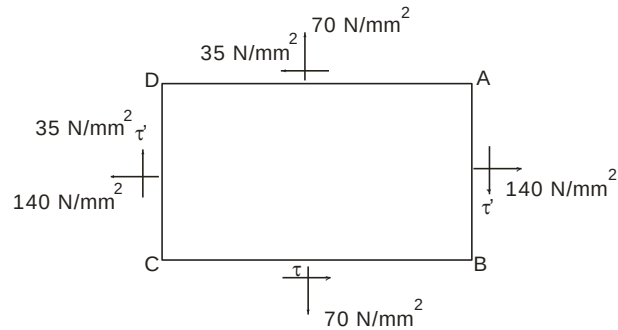


Fig. 9.20

Problem 9.13 At a point in a bracket the stresses on two mutually perpendicular planes are 240 N/mm^2 tensile and 120 N/mm^2 tensile. The shear stress across these planes is 60 N/mm^2 .

Find using the Mohr's stress circle, the principal stresses and maximum shear stress at the point.

Solution: Using Fig. 9.18 mark OA and OB to represent the normal stress of 240 N/mm^2 (tensile) and 120 N/mm^2 (tensile) respectively. Draw $AG = BF =$ shear stress of 60 N/mm^2 with C, the middle point of AB as centre, describe a circle with CF or CG as radius and obtain the points A and B.

Major principle stress = $\sigma_{np1} = OH = 264 \text{ N/mm}^2$ tensile (by measurement)

Minor principle stress = $\sigma_{np2} = OJ = 80 \text{ N/mm}^2$ tensile (by measurement)

Problem 9.14 A circle of 100 mm diameter is drawn on a mild steel plate before it is stressed as shown in Fig. 9.21. Find the lengths of the major and minor axes of the ellipse formed as a result of the deformation of the circle marked.

$$E = 2 \times 10^5 \text{ N/mm}^2 \text{ and } \frac{1}{m} = \frac{1}{4}$$

Solution:

$$\sigma_1 = 160 \text{ N/mm}^2$$

$$\sigma_2 = 40 \text{ N/mm}^2$$

$$\tau = 80 \text{ N/mm}^2$$

the principal stresses are given by ,

$$\frac{\sigma_1 + \sigma_2}{2} \pm \sqrt{\left(\frac{\sigma_1 - \sigma_2}{2}\right)^2 + \tau^2}$$

$$\frac{\sigma_1 + \sigma_2}{2} = \frac{160 + 40}{2} = 100 \text{ N/mm}^2$$

$$\frac{\sigma_1 - \sigma_2}{2} = \frac{160 - 40}{2} = 60 \text{ N/mm}^2$$

$\therefore$ The principal stresses are given by

$$100 \pm \sqrt{60^2 + 80^2} = 100 \pm 100 \text{ N/mm}^2$$

$\therefore$

$$\sigma_1 = 200 \text{ N/mm}^2 \text{ and } \sigma_2 = 0$$

$$e_1 = \frac{\sigma_1}{E} - \frac{\sigma_2}{mE} = \frac{200}{2 \times 10^5} = \frac{1}{1000}$$

$\therefore$ Increase in length of the diameter of the circle

$$= e_1 d = \frac{1}{1000} \times 100 = \frac{1}{10} \text{ mm} = 0.1 \text{ mm}$$

$$e_2 = \frac{P_2}{E} - \frac{P_1}{mE} = 0 - \frac{200}{4 \times 2 \times 10^5} = -\frac{1}{4000}$$

$\therefore$ Decrease on length of the diameter of the circle

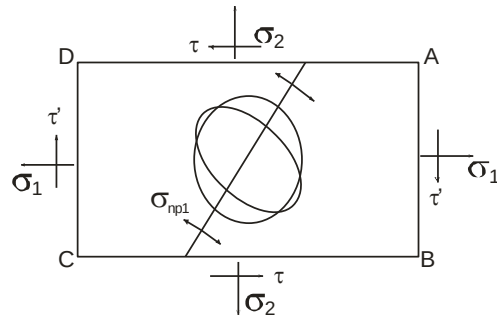


Fig. 9.21

$$e_1 d = -\frac{1}{4000} \times 100 = -\frac{1}{40} \text{ mm} = -0.025 \text{ mm}$$

Thus the circle will become an ellipse whose major axis

$$= 100 + 0.1 = 100.1 \text{ mm}$$

And minor axis $= 100 - 0.025 = 99.975 \text{ mm}$

Exercise

9.1 Define principal stresses and principal planes.

9.2 Write a note on Mohr's circle of stresses.

9.3 Define the term obliquity and how it is determined.

9.4 Derive an expression for the stresses on an oblique plane of a rectangular body when a body is subjected to simple shear.

9.5 Derive an expression for the principal stresses on an oblique plane of a rectangular body when a body is subjected to two mutually perpendicular stresses accompanied by simple shear.

9.6 Find the diameter of a circular bar which is subjected to an axial pull of 150 kN, if the maximum allowable shear stress on any section is 60 N/mm^2 .

(Ans. 39.89 mm)

9.7 The principal tensile stress at a point across two mutually perpendicular planes are 100 N/mm^2 and 50 N/mm^2 . Determine the normal and tangential stresses on a plane inclined at 30° to the axis of minor principal stress.

(Ans. 87.5 N/mm^2 , 21.65 N/mm^2)

9.8 At a point in a two dimensional system the normal stress on two mutually perpendicular planes are σ_1 and σ_2 and shear stress is τ . Show that one of the principal stresses is zero if $\tau = \sqrt{\sigma_1 \times \sigma_2}$.

9.9 A strained material is subjected to two dimensional stresses. Prove that the sum of the normal components of stresses on any two mutually perpendicular planes is constant.

9.10 A rectangular block of material is subjected to a tensile stress of 100 N/mm^2 on one plane and a tensile stress of 50 N/mm^2 on a plane at right angles, together with shear stresses of 60 N/mm^2 on the same planes. Find :

- (iv) The direction of the principal planes.
- (v) The magnitude of the principal stresses.
- (vi) The magnitude of the greatest shear stress.

(Ans. (i) $33^\circ 41'$ or $123^\circ 41'$ (ii) 140 N/mm^2 and 10 N/mm^2 , (iii) 65 N/mm^2)